Predefining an analysis plan

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Write the analysis decisions before inspecting primary outcomes

Preserve the original plan, denominator and visible reasons for later changes.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Write the analysis decisions before inspecting primary outcomes

> Preserve the original plan, denominator and visible reasons for later changes.

Learning goals

I can explain the **mechanism** and terminology of predefining an **analysis plan**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Privacy and data governance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain predefining an **analysis plan** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of predefining an **analysis plan**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Preserve the original plan, denominator and visible reasons for later changes.

Start with the research problem

An analysis plan fixes the question, primary outcome, eligible units, exclusions, missing-result handling, comparisons and uncertainty method before primary results are inspected. It reduces hidden flexibility; it does not make a weak method valid. If a later change is needed, retain the original plan and report what changed, why and when.

A label saying “v1” or a timestamp written into a document is not independent proof of its history.

Original concept explanation

An **analysis plan** records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected. **Deviation**s remain visible.

Complete case inputs, rules and limits

Plan item	Complete supplied exercise contract
Main population of cases	All twenty included by the original plan
Later proposed change	Drop four failed cases because they make the graph look worse
Known outcomes	Four proposed exclusions are failures; outcomes of the other sixteen are not supplied
Original denominator	Twenty; use actual pass count, not an invented sixteen passes
Picture rule	Twelve cases; exclude only unreadable files identified before scoring; report all versions
Picture change	Two readable model failures proposed for exclusion
Scope	Fictional audit; no actual timestamp, preregistration or experiment is created

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

Complete main-case evidence ledger

Main record	Supported statement	Unsupported shortcut
Original plan	All twenty cases included	Reconstruct eligibility after seeing outcomes
Four cases fail	Failure is known for those four	Assume every remaining case passed
Remove because graph looks worse	Outcome-informed exclusion violates original plan	Call it a predeclared eligibility rule
Original report	Actual passes / 20; numerator not supplied	Report 16/20 as an observed result
Conditional illustration	If all other sixteen passed, 16/20 = 80%	Treat the if-clause as evidence
Necessary revision	Retain original record and disclose revision/reason/timing	Erase old plan or present revised analysis as original

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Predefining an analysis plan — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse.	Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.	The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Exclude two readable cases because the new model fails them.

▼ **Check after predicting**

Prediction: That outcome-driven exclusion violates the recorded rule.

Reason: Model failure is not the predeclared unreadable-file criterion.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse.

The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

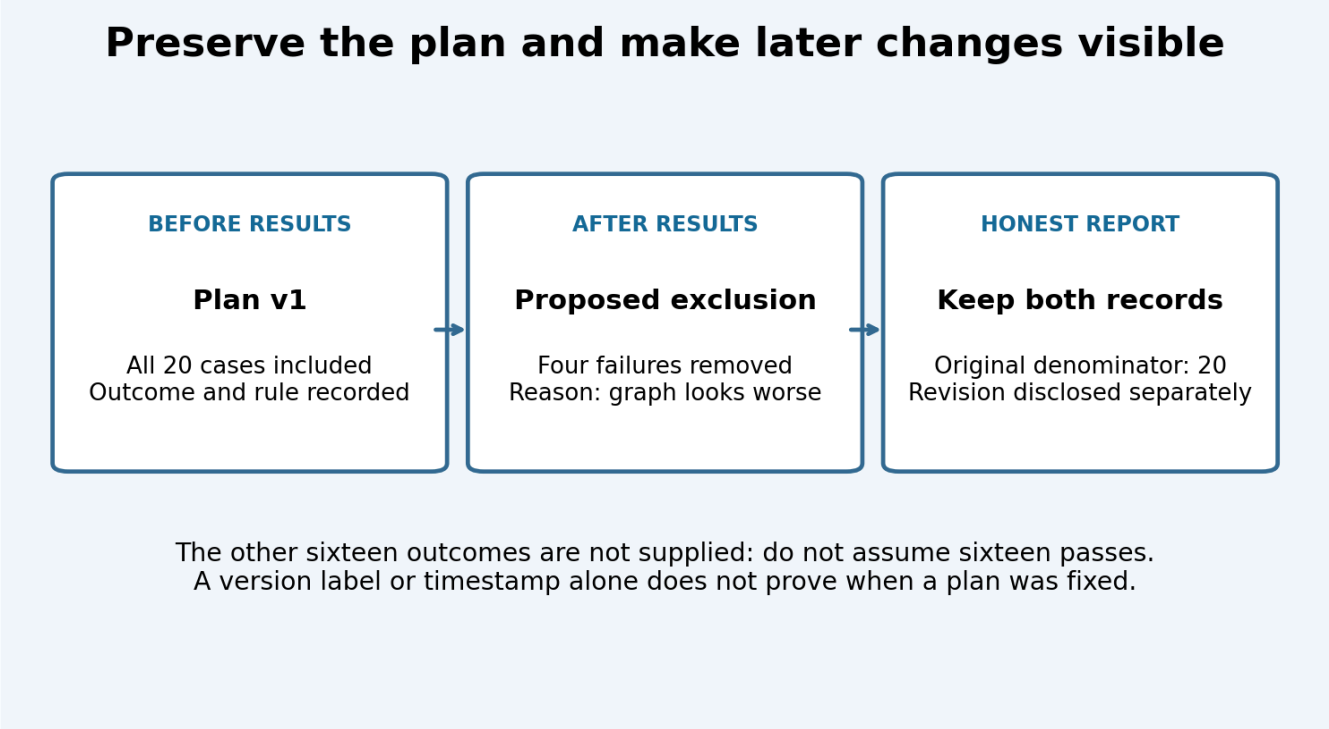
Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

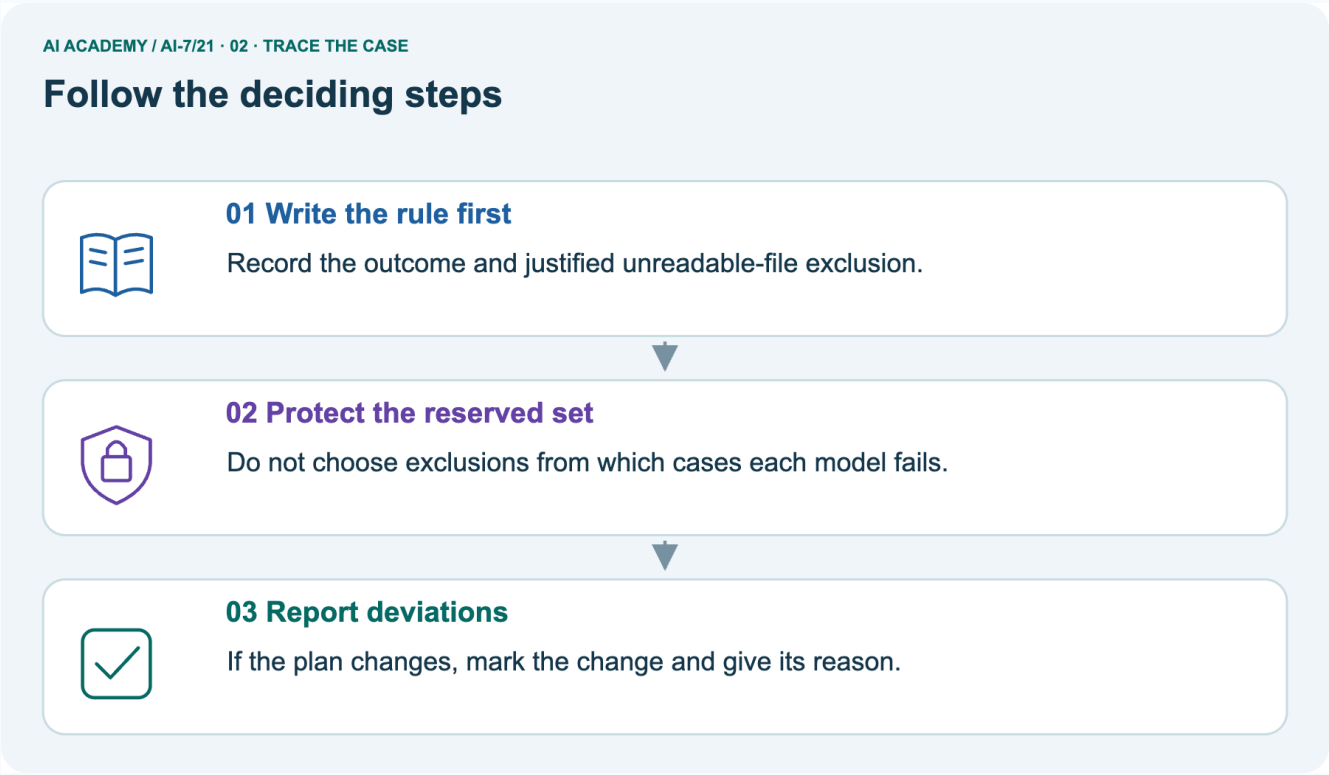
Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

The separate picture records a twelve-case plan before scoring. Only unreadable files identified before scoring are eligible for exclusion; all version results are reported. A readable case on which the model fails is not unreadable. Removing two such cases after seeing failures violates this rule.

The picture supplies no actual unreadable-file count or pass count, so no twelve-case performance rate can be calculated. A later necessary correction must preserve the original plan and analysis and be identified as a deviation, with reasons and timing. This fictional account does not prove a real plan was fixed before outcomes.

Read the labelled representation





Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the rule first	Record the outcome and justified unreadable-file exclusion.
2. Protect the reserved set	Do not choose exclusions from which cases each model fails.
3. Report deviations	If the plan changes, mark the change and give its reason.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

Complete picture evidence and limits

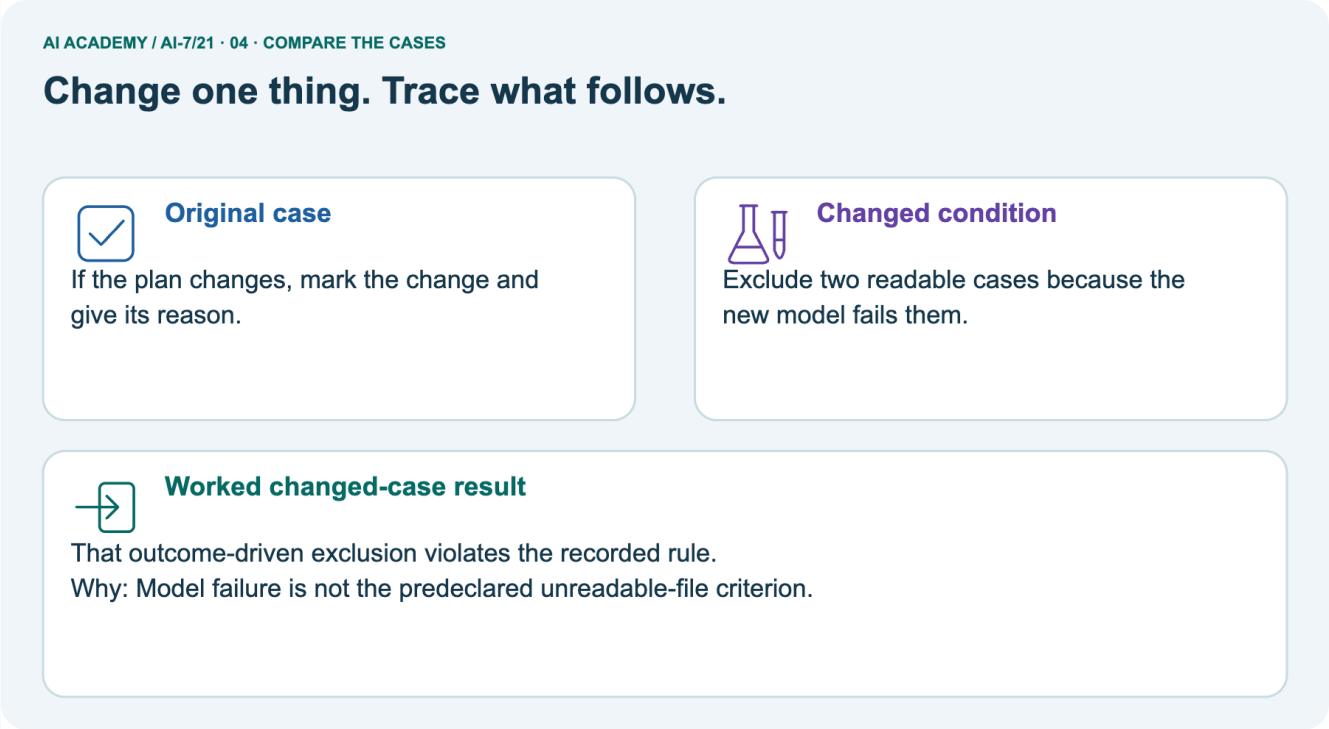
Picture evidence	Rule application
Twelve planned cases	Starting set, not a supplied twelve-case pass count
Unreadable before scoring	Exclusion allowed by this stated rule
Readable model failure	Does not meet unreadable-file exclusion
Two readable failures removed	Outcome-informed deviation; retain original analysis
Actual unreadable count / actual passes	Not supplied; no invented rate

Picture evidence	Rule application
Documented revision	Transparent change, not proof the original design was valid

Predict the changed condition

- Change one thing: Exclude two readable cases because the new model fails them. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

- Read the labelled picture: Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

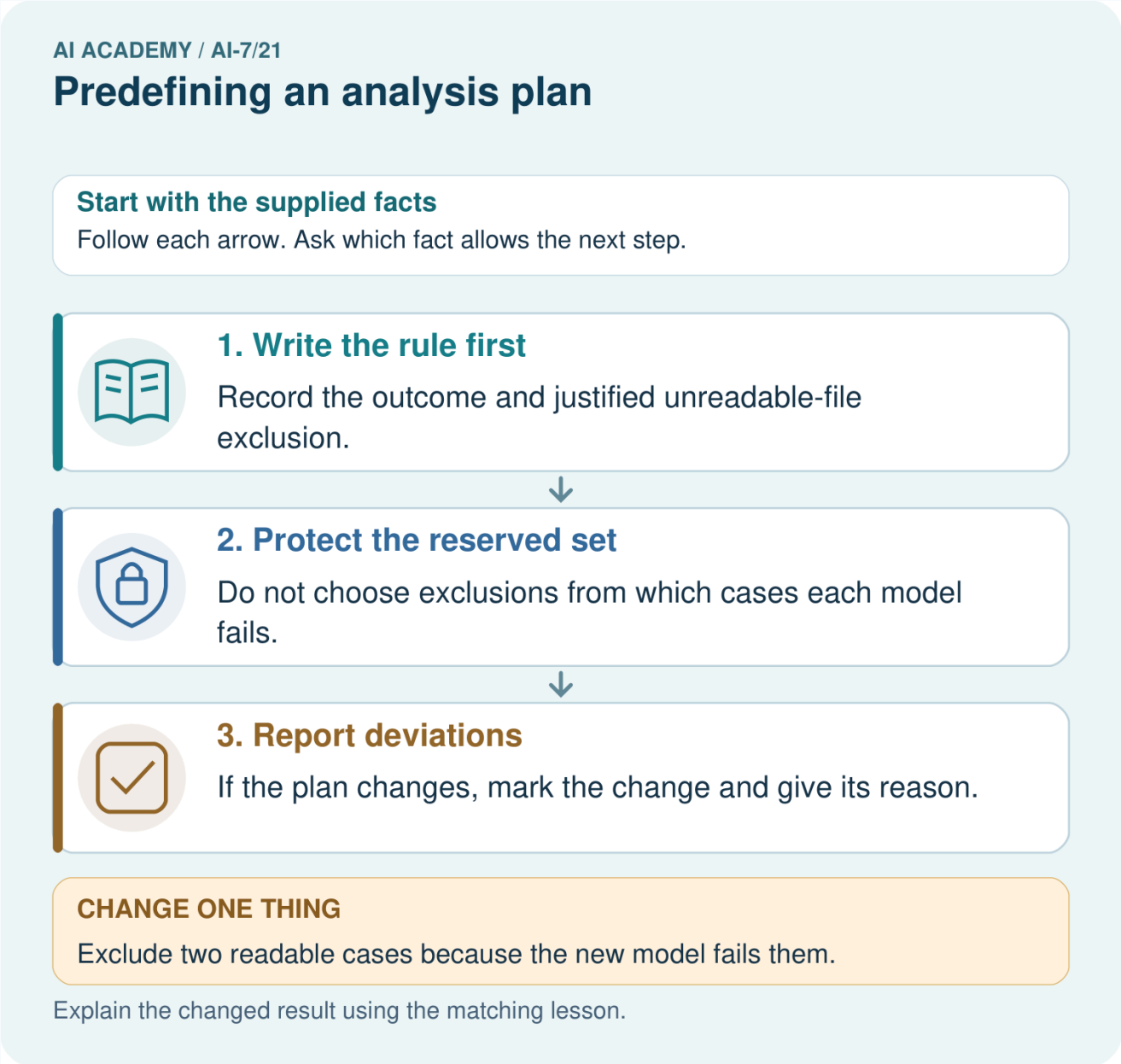
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Predefining an analysis plan

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Plan audit	Checked interpretation
All twenty cases planned	Four unattractive failures stay in original analysis
Sixteen other cases remain	Their pass/fail outcomes are unknown
Change disclosed	Makes change visible; does not remove all resulting bias
Version label or timestamp	Metadata alone does not prove chronology
Predeclared exclusion truly met	Report excluded IDs, rule and coverage
Failures are readable	Not eligible under unreadable-file rule
Analysis plan exists	Sampling and test assumptions still require justification

Write and justify the repair

- A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

> Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example “Predefining an **analysis plan**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Predefining an **analysis plan**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
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> _____

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Perform a paper comparison of the original plan and proposed exclusion. Record the original denominator twenty, four known failures and sixteen unspecified outcomes. Do not invent a final rate; write actual passes/20 and the conditional example separately. Then apply the picture's unreadable-before-scoring rule to the two readable failures.

   Preserve a hypothetical revision log with clearly labelled proposed reasons and timing, without claiming any real preregistration or study occurred. Reserve the fifteen-input independent task for the first individual attempt.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Use the supplied malformed-input rule, show eligible performance and report excluded-case coverage separately.

Complete independent task

Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Attempt this without hints

Plan excludes malformed **input**s by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

> **My independent procedure, result and limitation**
>
> | |
> |---|
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

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MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled predefining an **analysis plan evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of predefining an **analysis plan**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of predefining an **analysis plan**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Analysis plan	Declared procedures before inspecting primary results
Exclusion rule	A stated criterion for removing an ineligible case
Deviation	A recorded departure from the original plan

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Predefining an analysis plan — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse.	Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.	The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

By the end, I can	How I will show it
Explain the mechanism and terminology of predefining an analysis plan	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.
- Which labelled step matches this detail? If the plan changes, mark the change and give its reason.
- 1. Protect the reserved set
- 2. Write the rule first
- 3. Report **deviation**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Report **deviation**s If the plan changes, mark the change and give its reason. An **analysis plan** limits flexibility after outcomes are known.

Necessary changes can be transparent, but they should not be silently presented as the original plan.

Changed-case question and optional hint

- Changed-case practice: Exclude two readable cases because the new model fails them. Which conclusion is supported by the supplied facts?
- 1. That outcome-driven exclusion violates the recorded rule.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Repair and retain the record

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

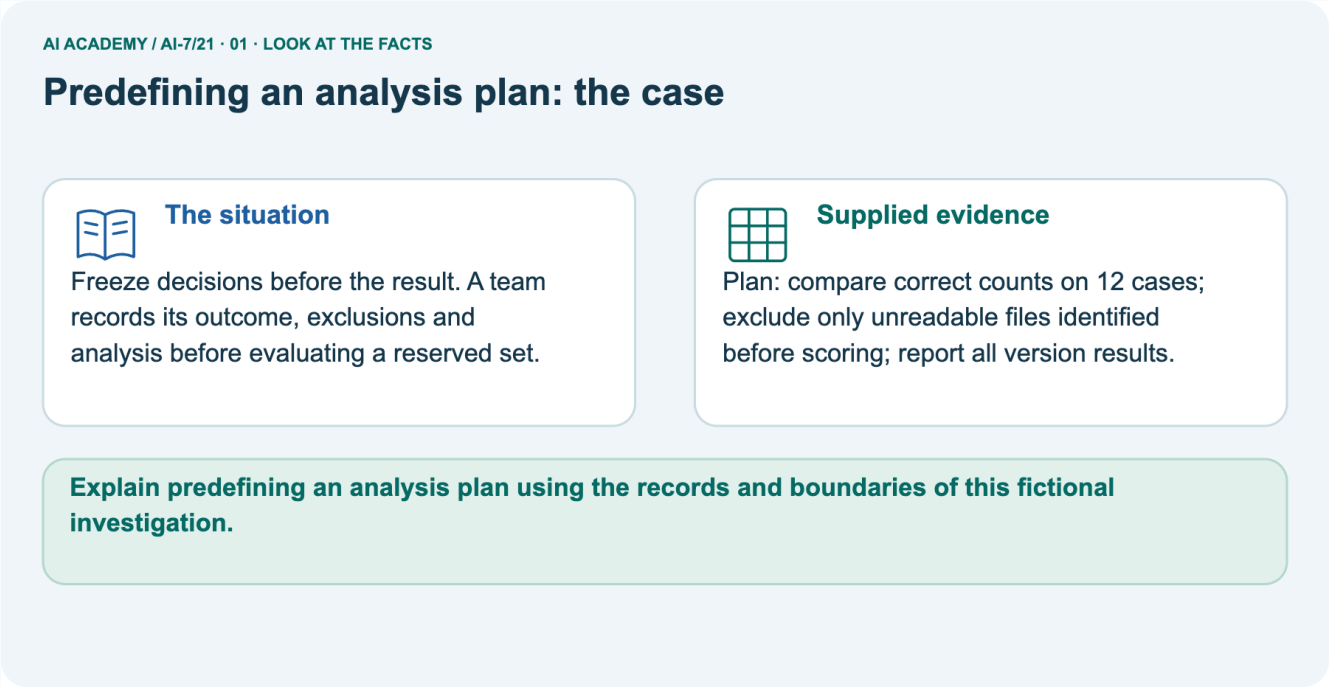
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Predefining an analysis plan: the case



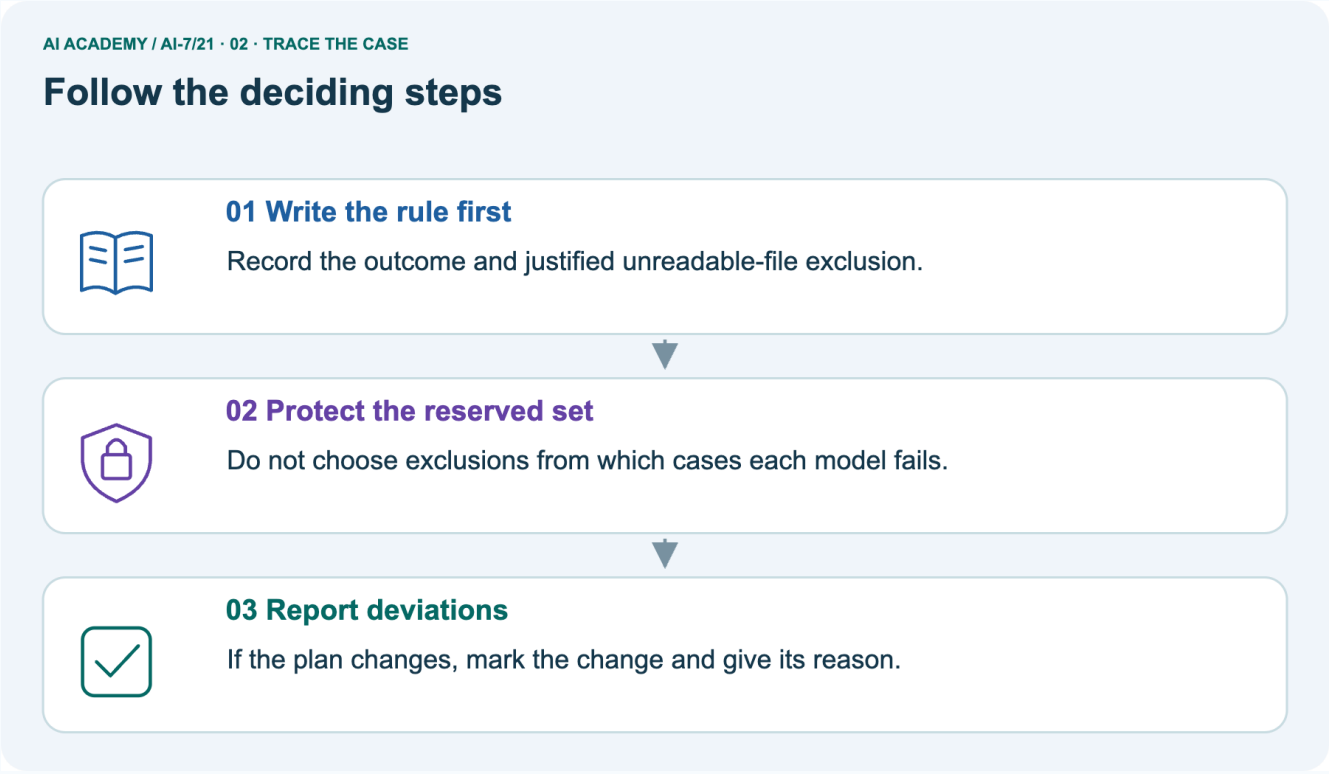
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Supplied facts: Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

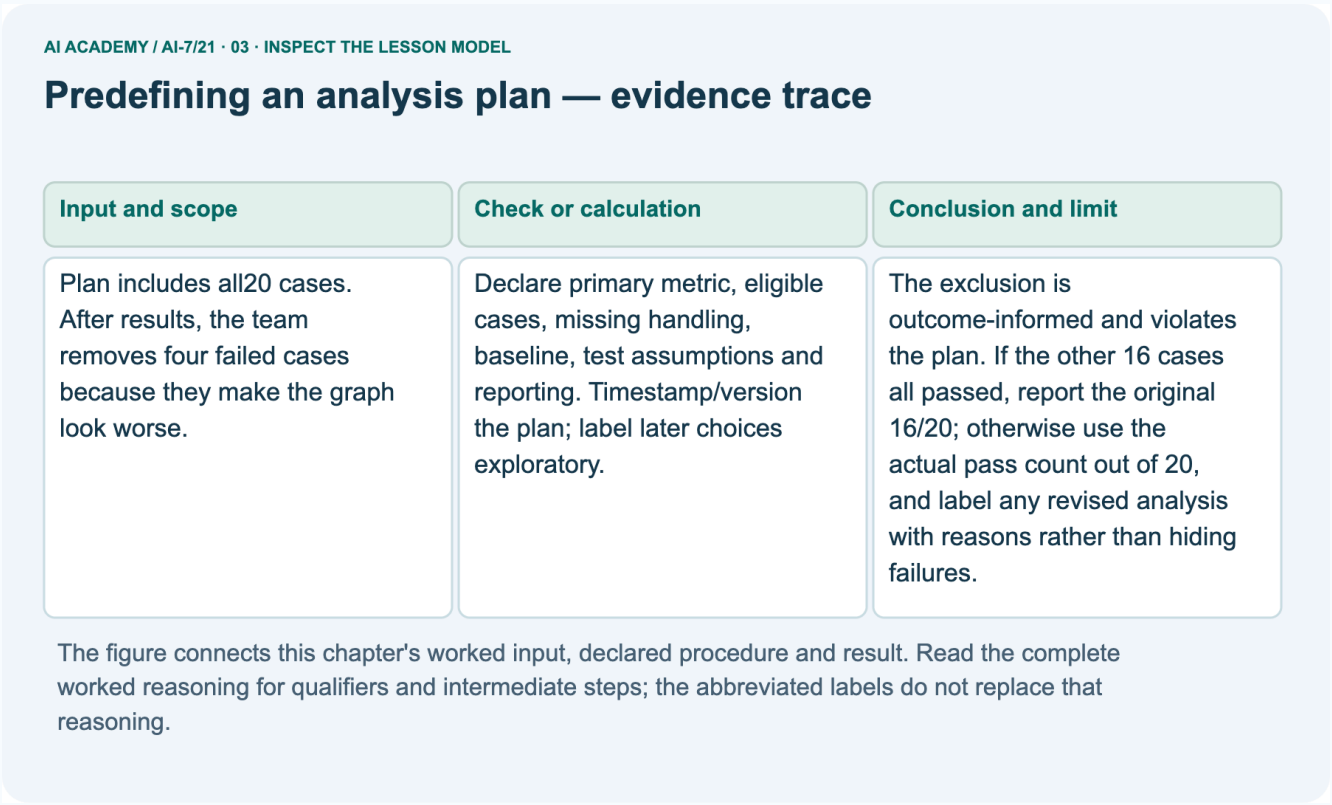


Follow the labelled picture with the matching explanation. Open the vector diagram

Write the rule first: Record the outcome and justified unreadable-file exclusion. → Protect the reserved set: Do not choose exclusions from which cases each model fails. → Report **deviation**s: If the plan changes, mark the change and give its reason.

Point to each numbered stage and explain which recorded **input** it uses.

03. Predefining an analysis plan — evidence trace

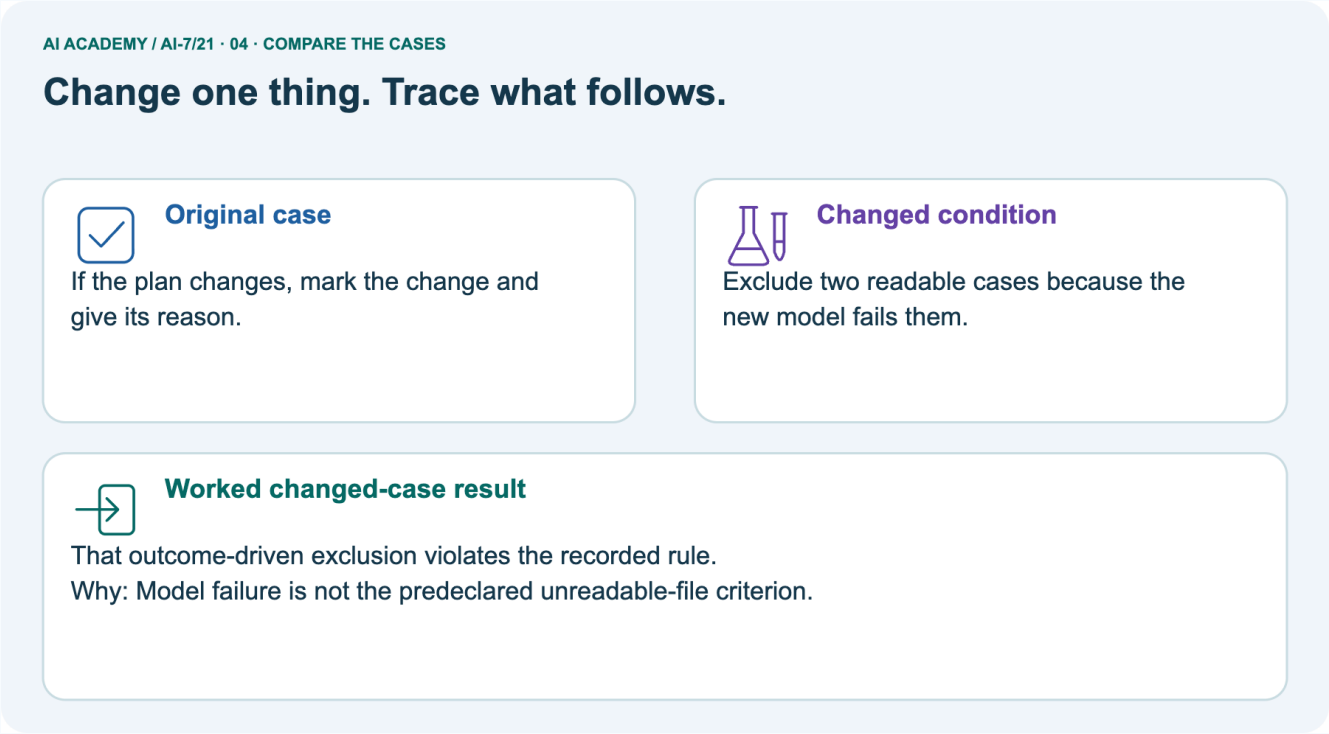


Follow the labelled picture with the matching explanation. Open the vector diagram

Predefining an **analysis plan** — **evidence** trace. Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. | Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory. | The exclusion is outcome-informed and violates the plan. If the other 16 cases all passed, report the original 16/20; otherwise use the actual pass count out of 20, and label any revised analysis with reasons rather than hiding failures.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



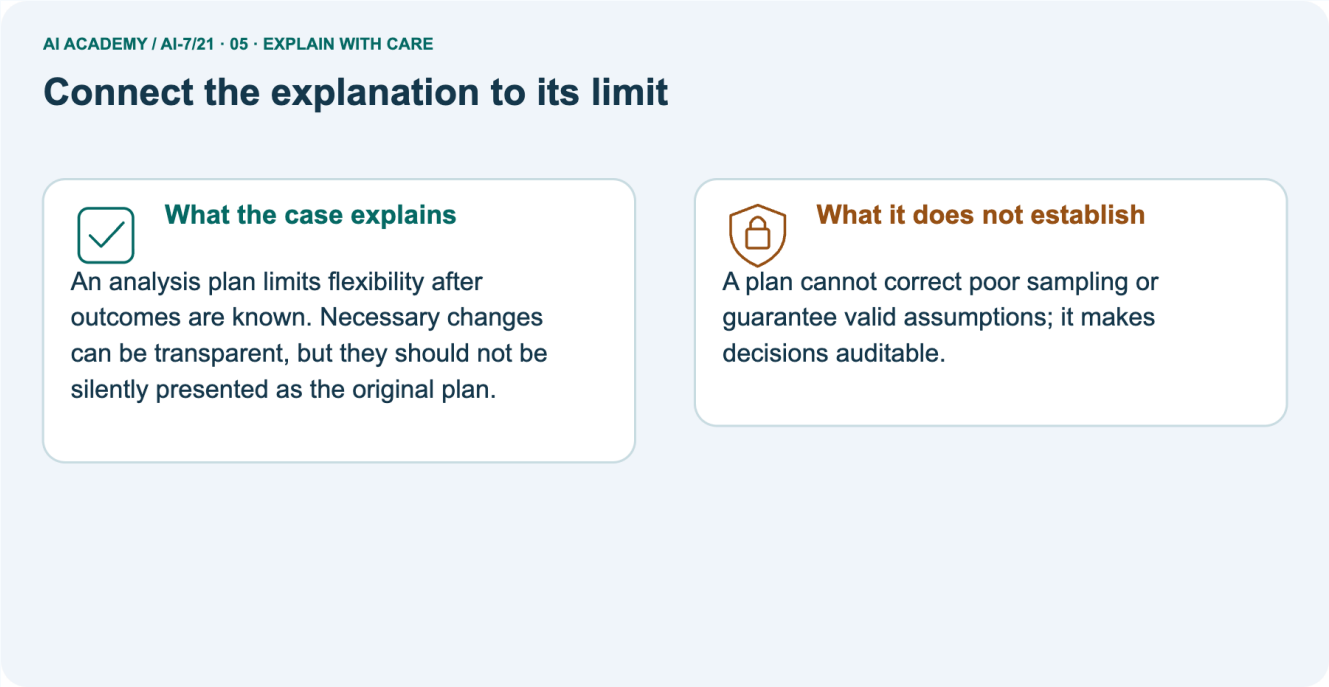
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the plan changes, mark the change and give its reason. Changed condition: Exclude two readable cases because the new model fails them.

Expected result: That outcome-driven exclusion violates the recorded rule. Reason: Model failure is not the predeclared unreadable-file criterion.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



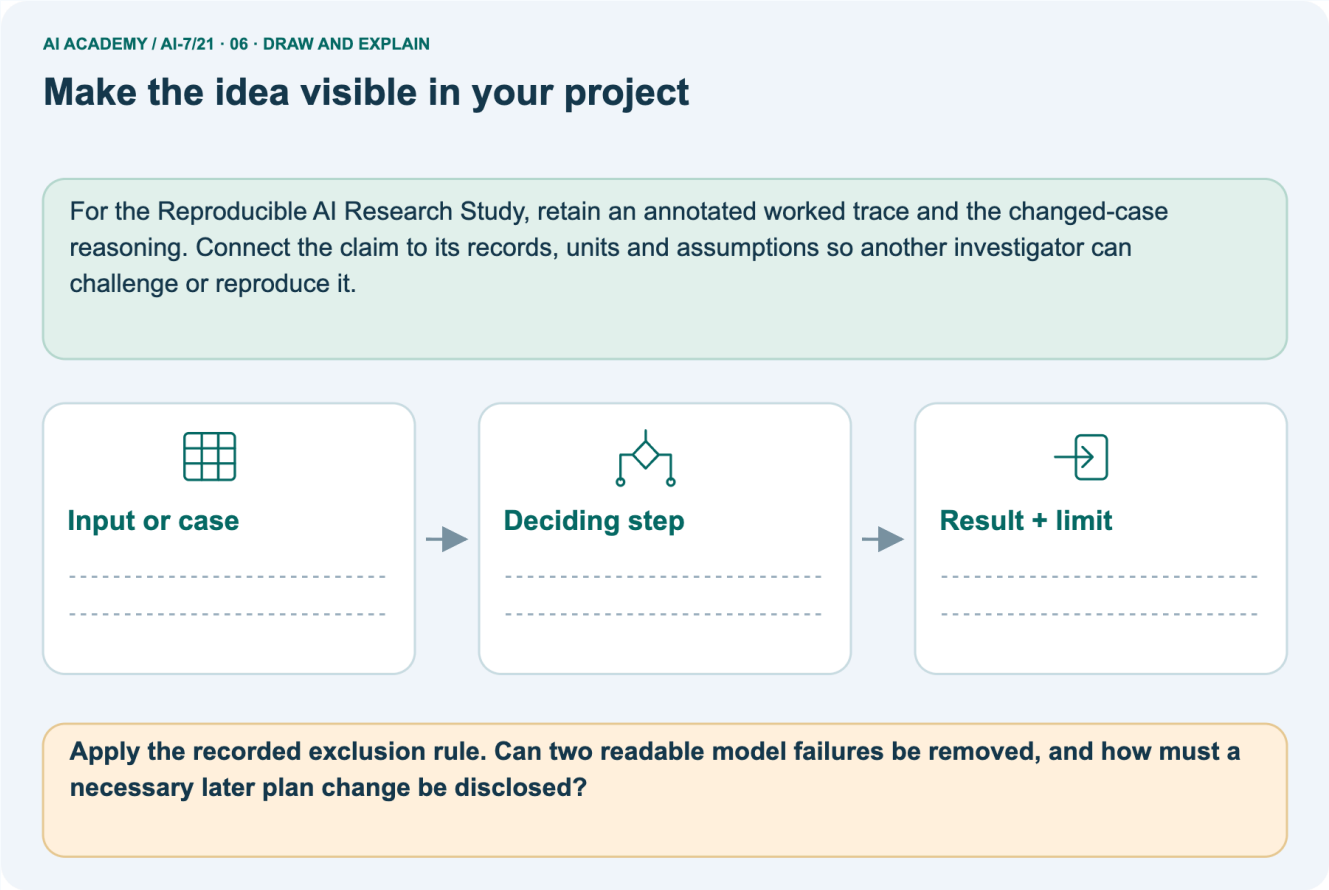
Follow the labelled picture with the matching explanation. Open the vector diagram

An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

Limit: A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Predefining an analysis plan: the case

02. Follow the deciding steps

03. Predefining an analysis plan — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.